Amy Wang

CONTACT INFORMATION

Personal E-mail: amywangsci@gmail.com Work E-mail: wanga84@gene.com
Website: https://amywangsci.github.io

PROFESSIONAL SUMMARY

Machine learning scientist specializing in generative models, LLM evaluation, and agentic AI systems for protein and antibody design. Deployed AI-driven discovery workflows now used across the large molecule portfolio at Genentech.

RESEARCH EXPERIENCE

Genentech, Prescient Design South San Francisco, CA

Senior ML Scientist II, Drug Discovery & Foundation Models Sept 2025 – Present

- Designed a tool-augmented LLM agent for antibody discovery, enabling autonomous sequence selection via structured tool calls
- Developed protein design tasks for LLM post-training and evaluated reasoning consistency and alignment with experimental outcomes
- Filed 2 patents on (1) generative AI-powered developability optimization and (2) interpretable ML affinity prediction that enables protein design and structural insights from sequence

Senior ML Scientist I, Large Molecule Drug Discovery Mar 2023 – Sept 2025

- Led AI strategy for 3 antibody programs, developing a sequence exploration approach now used organization-wide to improve PK and aggregation
- Led research on dataset scaling and affinity maturation in external collaboration, applying Bayesian optimization for efficient sequence exploration
- Developed a generative sequence model that distilled antibody biophysical constraints into scalable priors, enabling efficient multi-property optimization
- Established model adoption standards through benchmarking and curation of pipeline-relevant training datasets
- Communicated findings through 1 arXiv preprint, 2 conference proceedings, and 3 workshop papers

Microsoft Research New England Cambridge, MA (*Virtual*)

BioML Research Intern Jun – Sept 2022

Advisors: Kevin Yang, Ava Amini & Alex Lu

- Conceptualized and implemented integration of biophysical features into graph neural networks for protein property prediction
- Presented findings at *NeurIPS Machine Learning for Structural Biology* Workshop showing Rosetta computed features improve out-of-distribution prediction on two benchmark tasks

Stanford University Stanford, CA

Graduate Student Researcher Apr 2019 – March 2023

Advisors: William Weis & Alexander Dunn

- Characterized key structural determinants of force-dependent cell adhesion mechanisms using single molecule force spectroscopy (SMFS) and quantitative biochemical assays
- Developed theoretical framework explaining how cooperative molecular interactions generate catch bonds, advancing understanding of adhesion biology mechanics in physiological shear conditions
- Devised a pipeline which increased data collection and analysis throughput by 10x
- Built a custom single-channel epi-fluorescent microscope

Stanford University Stanford, CA
Rotation Student Jan – Apr 2019
Advisor: Ron Dror
- Conducted molecular dynamics simulations of membrane protein ion transport, contributing to findings published in Nature

Education

Stanford University, Stanford, CA 2018-2023
Ph.D., Chemical Engineering
Thesis: Mechanisms governing the force-dependent interactions between αE-catenin and F-actin
Advisors: William Weis (Depts. Structural Biology, Molecular and Cellular Physiology)
Alexander Dunn (Dept. Chemical Engineering, Biophysics Program)

Massachusetts Institute of Technology, Cambridge, MA 2014-2018
Bachelor of Science in Chemical Engineering, Minor in Polymers and Soft Matter

Publications and Preprints

N.C. Frey, I. Hötzel, S.D. Stanton, R. Kelly, R.G. Alberstein, ..., **A. Wang**, ..., A. Regev, Y. Wu, K. Cho, R. Bonneau, V. Gligorijević. Lab-in-the-loop therapeutic antibody design with deep learning. *bioRxiv* 2025. [Link]

A.A. Ismail, T. Oikarinen, **A. Wang**, J. Adebayo, S.D. Stanton, H.C. Bravo, K. Cho, N.C. Frey. Concept Bottleneck Language Models For Protein Design. *International Conference on Learning Representations (ICLR)* 2025. [Link]

C. Wang, M. Uehara, Y. He, **A. Wang**, A. Lal, T. Jaakola, S. Levine, A. Regev, H. Wang, T. Biancalani. Fine-tuning discrete diffusion models via reward optimization with applications to DNA and protein design. *International Conference on Learning Representations (ICLR)* 2025. [Link]

NA Bax*, **A Wang***, DL Huang, S Pokutta, AR Dunn, WI Weis. Multi-level force-dependent allosteric enhancement of αE-catenin binding to F-actin by vinculin. *Journal of Molecular Biology* 2023. [Link]

A Wang, AR Dunn, WI Weis. Mechanism of the cadherin-catenin F-actin catch bond interaction. *eLife* 2022. [Link]

S Bose, LR Volpatti*, D Thiono*, V Yesilyurt, C McGladrigan, Y Tang, A Facklam, **A Wang**, S Jhunjhunwala, O Veiseh, J Hollister-Lock, C Bhattacharya, GC Weir, DL Greiner, R Langer, DG Anderson. A retrievable implant for the long-term encapsulation and survival of therapeutic xenogeneic cells. *Nat. Biomed. Eng.* 2020. [Link]

TA Chew*, BJ Orlando*, J Zhang*, NR Latorraca, **A Wang**, SA Hollingsworth, DH Chen, RO Dror, M Liao, L Feng. Structure and mechanism of the cation–chloride cotransporter NKCC1. *Nature* 2019 572:488–492. [Link]

A Huang, JM Paloni, **A Wang**, AC Obermeyer, HV Sureka, H Yao, BD Olsen. Predicting protein–polymer block copolymer self-assembly from protein properties. *Biomacromolecules* 2019, 20, 10, 3713-3723. [Link]

S Chopra, N Bertrand, J Lim, **A Wang**, O Farokhzad, R Karnik. Design of insulin-loaded nanoparticles enabled by multistep control of nanoprecipitation and zinc chelation. *ACS Applied Materials & Interfaces* 2017 9 (13), 11440-11450. [Link]

Workshop Papers

A Wang, Z. Sang, S.D. Stanton, J.L. Hofmann, [...], N.C. Frey, A.M. Watkins, F. Seeger. A guided design framework for the optimization of therapeutic-like antibodies. *Generative and Experimental Perspectives for Biomolecular Design, ICLR* 2025.

P. Rao, H. Isaacson, J.L. Hofmann, D. Davidson, **A. Wang**, A.M. Watkins, R. Bonneau, S. Izadi, J.H. Lee. SurfProp: A surface-based property prediction framework for antibody developability and screening *GenBio Workshop, ICML* 2025.

Z. Ma, D. Davidson, **A Wang**, N. Frey, F. Seeger. The effects of structural conditioning on antibody inverse folding. *Women in ML Workshop, NeurIPS* 2023.

A Wang, AP Amini, AX Lu, KK Yang. Learning from physics-based features improves protein property prediction. *Machine Learning in Structural Biology Workshop, NeurIPS* 2022. [Link]

Honors and Awards

University Nominee for the Schmidt Science Fellows *8 Stanford students nominated*	2022
Valuation of Public Companies in the Life Sciences Pitch Competition - First Place	2020
Stanford Graduate Fellowship in Science & Engineering	2018-2023
NSF Graduate Research Fellowship	2018-2023
ChemH Chemistry/Biology Interface Predoctoral Training Program	2018-2023
Tau Beta Pi Honors Society	2017
MIT Chemical Engineering Departmental BP Academic Achievement Award	2016

Service and Leadership

Stanford Graduate Life Office, Head Community Associate	2021-2023
Stanford Molecular & Cellular Physiology DEI Committee Liaison	2020-2022
Stanford Graduate Life Office, Community Associate	2019-2021
Bay Area Graduate Pathways to STEM Mentor & Panel Speaker	2018-2020

Skills

Machine Learning
Scientific reasoning evaluation, tool-augmented LLM agents, sequence- and text-based modeling, Bayesian optimization, model interpretability

Biophysical Modeling
Protein surface representations, PyRosetta, AMBER molecular dynamics, PyMOL